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A Robust Patch-Based Hierarchical Framework for Image Forgery Detection and Classification

23CSE498 – Project Phase 2 (Panel Review 1)

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Abstract

Digital image manipulation has become increasingly sophisticated due to modern editing software, generative AI synthesis, and deep learning techniques. Visually seamless manipulations are exceptionally difficult to identify using conventional image analysis because tampering artifacts are localized, subtle, and frequently concealed by post-processing operations such as JPEG recompression, Gaussian blurring, sensor noise injection, and rescaling.

This project proposes a robust patch-based hierarchical framework for image forgery detection and classification. The framework processes every image patch through three complementary forensic streams: a spatial stream based on raw RGB pixel projections, a frequency stream based on the 2D Fast Fourier Transform (FFT) magnitude spectrum, and a noise-residual stream based on high-pass filtering (HPF). The resulting representations are projected into a common token space and adaptively combined using a cross-attention fusion layer before global reasoning by a 12-layer Vision Transformer (ViT) encoder. A two-stage hierarchical decision strategy first determines whether an image is authentic or tampered, and subsequently identifies the manipulation category as Copy-Move, Splicing, or Inpainting.

During the current implementation phase, the core feature extraction, three-stream tokenization, cross-attention fusion, ViT backbone, data pipeline, evaluation, and explainability modules have been fully integrated. Preliminary evaluation on a held-out benchmark test set (≈2,700 images) reports 88.95% accuracy and 92.14% F_1-score for binary forgery detection, while multi-class forgery-type classification currently achieves 67.02% accuracy and 69.30% macro F_1-score. Current optimization focuses on mitigating Copy-Move/Inpainting confusion via hard-negative mining and class-weighted loss.

1 Problem Statement & Motivation

1.1 Problem Context

With the rapid expansion of digital media, social networks, and generative AI tools, the integrity of digital imagery has become paramount. Malicious actors can alter visual evidence with high precision, creating convincing forgeries that fool human observers and conventional verification systems. The project focuses on three major image manipulation categories:

- **Copy-Move:** A region is duplicated and pasted elsewhere within the same image to conceal or replicate objects.

- **Splicing:** A region originating from an external source image is composited into the target image.
- **Inpainting:** An object or region is removed and the missing void is synthetically hallucinated using generative methods.

1.2 Core Forensic Challenges

Forensic traces rarely manifest uniformly across all images. While tampering may appear visually seamless in the spatial RGB domain, it frequently disrupts underlying physical camera properties, such as photo-response non-uniformity (PRNU) and periodic JPEG blocking artifacts in the frequency domain. Furthermore, real-world images undergo heavy post-processing (compression, blurring, downsampling), which intentionally or unintentionally degrades pixel-level edge artifacts.

1.3 Problem Statement

Digital image manipulations such as Copy-Move, Splicing, and Inpainting are difficult to accurately identify from digital images because modern editing produces visually seamless appearances with minimal perceptible spatial artifacts. Existing forensic datasets like CASIA and DEFACTO contain diverse manipulation characteristics, resolutions, and compression levels, making it challenging for conventional single-domain models to consistently detect and classify forgeries across different conditions.

Therefore, this project aims to develop a robust patch-based hierarchical framework using a **Three-Stream Vision Transformer (ViT)** that jointly processes spatial, frequency (2D FFT), and noise-residual tokens via cross-attention. By structuring the pipeline into binary tamper detection followed by forgery-type classification, the framework aims to eliminate class confusion on authentic images, improve cross-dataset generalization, and remain robust against real-world post-processing attacks.

Formally, given a digital query image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, the framework solves two hierarchical tasks:

1. **Stage 1 (Binary Detection $\mathcal{T}_{\text{det}}$):** Predict $y_{\text{det}} \in \{0, 1\}$ indicating whether $\mathbf{I}$ is Authentic (0) or Tampered (1).
2. **Stage 2 (Forgery Classification $\mathcal{T}_{\text{cls}}$):** Conditioned on tampering ($y_{\text{det}} = 1$), predict $y_{\text{cls}} \in \{\text{Copy-Move}, \text{Splicing}, \text{Inp}$

2 Literature Survey & Research Gap

2.1 Literature Analysis

- **Block/Keypoint Methods (Fridrich, 2003):** Effective for simple copy-move detection but highly sensitive to geometric transforms (rotation, scaling) and fail completely in smooth, textureless regions.
- **Global CNN Models:** Capture semantic scene context well, but receptive field pooling causes them to miss fine, localized tampering artifacts.
- **Patch-Based CNNs:** Improve local boundary analysis but remain confined strictly to the spatial domain, ignoring frequency and noise anomalies.
- **Standard Vision Transformers (Dosovitskiy et al., 2021):** Use single-stream spatial RGB tokens, ignoring forensic noise and spectral cues.
- **Frequency/Noise Methods:** Analyze DCT/FFT or noise residuals independently as preprocessing inputs, failing to learn joint cross-domain token representations.

2.2 Comparative Baseline Analysis

Table 1 outlines the limitations of existing state-of-the-art methods and illustrates how the proposed architecture fills these gaps.

Paper / Method	Primary Task	Supervision Type	Major Limitation
Block-based (Fridrich, 2003)	Copy-Move Detection	Block matching	Sensitive to rotation, scaling, and JPEG compression
Frequency-aware CNN	Forgery Detection	Image-level labels	Frequency used only as fixed preprocessing; no joint learning
Standard ViT (ICLR 2021)	Image Classification	Image-level labels	Uses only spatial tokens; ignores forensic noise and spectral cues
ManTra-Net (CVPR 2019)	Localization	Pixel-level supervision	Feature concatenation only; lacks unified multi-domain reasoning
MVSS-Net (TPAMI 2022)	Detection + Localization	Pixel-level supervision	Multi-view CNN; lacks frequency tokens and Transformer reasoning
TruFor (CVPR 2023)	Detection + Localization	Pixel-level supervision	Two-stream (RGB + noise); lacks frequency stream and token fusion
Proposed Three-Stream ViT	**Detection + Classification**	**Hierarchical Image Labels**	**Addresses limitation: Joint spatial, frequency, and noise reasoning via cross-attention and ViT backbone**

Table 1: Comparison with existing forgery detection literature and baseline methods.

2.3 Identified Research Gap

No existing work fuses spatial RGB, 2D FFT frequency, and noise-residual patch representations as parallel token streams within a Vision Transformer. The proposed architecture treats all three representations as first-class tokens and combines them via cross-attention before global Transformer reasoning.

3 Originality & Proposed Contribution

The principal contributions of this work are summarized as follows:

1. **Tri-Stream Patch Tokenization:** Each patch is simultaneously encoded into spatial, spectral, and noise-residual embedding tokens, capturing complementary forensic fingerprints.
2. **Cross-Attention Fusion Layer:** Rather than naive feature concatenation, a cross-attention layer learns adaptive inter-domain dependencies, dynamically emphasizing the domain that reveals the strongest tamper evidence.
3. **Global ViT Backbone Reasoning:** Leveraging a 12-layer Vision Transformer encoder allows the model to compute global multi-head self-attention across all patches, enabling the detection of widely separated cloned regions in Copy-Move attacks.
4. **Two-Stage Masked Hierarchical Strategy:** Incorporates a gated loss function where authentic images supervise only binary detection, preventing unmanipulated images from corrupting the multi-class forgery attribution boundary.
5. **Post-Processing Defense:** Incorporates on-the-fly JPEG compression, Gaussian blur, sensor noise, and rescaling augmentations to guarantee real-world generalization.

4 Proposed Methodology

4.1 Overall Pipeline

The pipeline executes through the following stages:

1. An input image $\mathbf{I} \in \mathbb{R}^{224\times224\times3}$ is divided into $N = 196$ non-overlapping patches of size 16×16.
2. Each patch p_i is simultaneously routed through three parallel feature extractors: raw RGB flattening, 2D FFT magnitude calculation, and Gaussian high-pass filtering.
3. Individual stream representations are linearly projected into a common embedding space $\mathbb{R}^d$.
4. A cross-attention layer fuses the three tokens into a unified forensic token $\mathbf{z}_i^* \in \mathbb{R}^d$.
5. Fused tokens are processed by a standard 12-layer Vision Transformer (ViT) encoder.
6. Output representations are fed into two task-specific heads: a binary detection head and a multi-class forgery-type classification head.

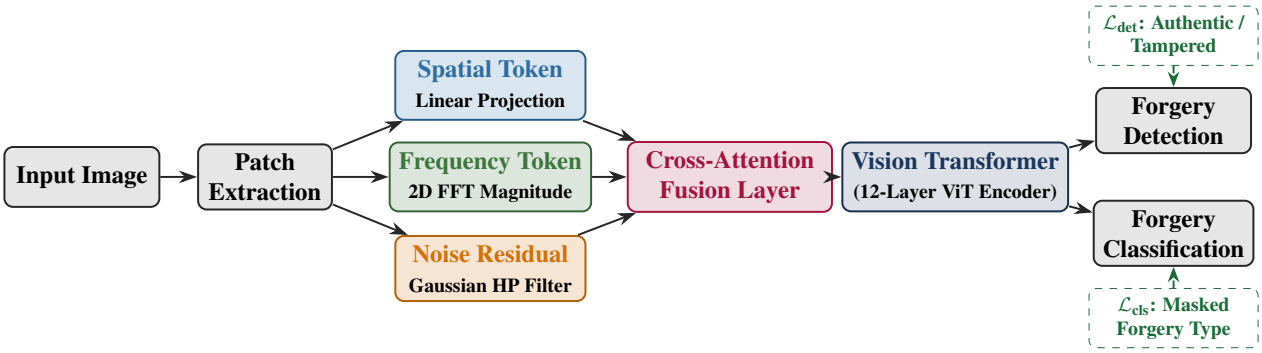


Figure 1: Proposed Three-Stream Tokenization Architecture with Vision Transformer (ViT) backbone and hierarchical supervision.

5 Mathematical Formulation

Let an image $\mathbf{I}$ be partitioned into non-overlapping patches $\{p_1, p_2, \ldots, p_N\}$, where $p_i \in \mathbb{R}^{P^2 \times C}$ ($P = 16, C = 3$).

5.1 Patch Tokenization

For each patch p_i, three distinct forensic tokens are generated:

1. **Spatial Token ($\mathbf{z}_i^s$):** Captures raw photometric textures, boundary discontinuities, and color inconsistencies via linear projection:

$$\mathbf{z}_i^s = W_s p_i + b_s, \quad W_s \in \mathbb{R}^{d \times (P^2 \cdot C)}. \tag{1}$$

2. **Frequency Token ($\mathbf{z}_i^f$):** Captures periodic JPEG blocking grid artifacts and high-frequency spectral anomalies via 2D Fast Fourier Transform magnitude:

$$\mathbf{z}_i^f = W_f \cdot \log\left(1 + |\mathcal{F}(p_i)|\right) + b_f, \tag{2}$$

   where $\mathcal{F}$ represents the 2D FFT and $|\cdot|$ extracts spectral magnitude.

3. **Noise-Residual Token ($\mathbf{z}_i^n$):** Suppresses low-frequency image semantics and isolates sensor noise (PRNU) patterns using a Gaussian High-Pass Filter (HPF):

$$\mathbf{z}_i^n = W_n \cdot (p_i - \text{Blur}_\sigma(p_i)) + b_n. \tag{3}$$

All three tokens are projected into a common latent embedding space of dimension $d = 768$.

5.2 Cross-Attention Fusion Layer

For patch i, the tokens are arranged as $\mathbf{T}_i = [\mathbf{z}_i^s, \mathbf{z}_i^f, \mathbf{z}_i^n]^T \in \mathbb{R}^{3 \times d}$. Treating the spatial token $\mathbf{z}_i^s$ as Query (Q) and the stacked multi-domain tokens as Key (K) and Value (V):

$$\mathbf{z}_i^* = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \in \mathbb{R}^d. \tag{4}$$

This allows the network to dynamically assign higher attention weights to frequency or noise tokens when spatial boundaries have been intentionally blurred.

6 Vision Transformer (ViT) Backbone

The sequence of fused forensic patch tokens $\mathbf{Z}^* = [\mathbf{z}_1^*, \mathbf{z}_2^*, \ldots, \mathbf{z}_N^*]$ is prepended with a learnable classification token $\mathbf{z}_{\text{cls}}$ and augmented with standard 1D learnable position embeddings $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+1) \times d}$:

$$\mathbf{Z}_0 = [\mathbf{z}_{\text{cls}}; \mathbf{z}_1^*; \mathbf{z}_2^*; \ldots; \mathbf{z}_N^*] + \mathbf{E}_{\text{pos}}. \tag{5}$$

6.1 Why Vision Transformer (ViT)?

Standard CNNs rely on local receptive fields that expand only gradually across deep layers, restricting their ability to compare distant image patches. The ViT backbone offers critical advantages for image forensics:

- **Global Multi-Head Self-Attention (MSA):** In Copy-Move forgeries, a region is copied and pasted elsewhere—often separated by large pixel distances. ViT computes full pairwise attention across all patches at every layer, enabling immediate matching of cloned regions regardless of spatial separation.
- **Consistent Sequence Modeling:** The fused tokens $[\mathbf{z}_i^*]$ seamlessly integrate into standard Transformer encoder blocks without requiring spatial downsampling or multi-resolution window constraints.
- **Architecture Details:** Uses a 12-layer Transformer encoder with 12 attention heads ($d = 768$, MLP dimension = 3072), Layer Normalization (LN), and residual connections.

7 Hierarchical Prediction & Loss Objective

7.1 Two-Stage Decision Mechanism

- **Stage 1 – Binary Forgery Detection:** Uses the output representation to predict whether the image is Authentic ($y_{\text{det}} = 0$) or Tampered ($y_{\text{det}} = 1$) via Binary Cross-Entropy loss $\mathcal{L}_{\text{det}}$.
- **Stage 2 – Forgery Classification:** Conditioned strictly on an image being tampered, a multi-class linear head predicts the specific tampering type: Copy-Move, Splicing, or Inpainting via Cross-Entropy loss $\mathcal{L}_{\text{cls}}$.

7.2 Hierarchical Masked Loss

A naive multi-task loss penalizes authentic images on the classification head, even though authentic images have no meaningful forgery category. To prevent corrupting decision boundaries, a binary indicator mask $\mathbb{I}(y_{\text{det}} = 1)$ is incorporated:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{det}}(\hat{y}_{\text{det}}, y_{\text{det}}) + \lambda \cdot \mathbb{I}(y_{\text{det}} = 1) \cdot \mathcal{L}_{\text{cls}}(\hat{y}_{\text{cls}}, y_{\text{cls}}) + \gamma \|\Theta\|_2^2, \tag{6}$$

where:

- $\lambda = 0.5$ balances the gradient updates between binary detection and forgery classification;
- $\mathbb{I}(y_{\text{det}} = 1)$ activates classification loss exclusively for tampered images; and
- $\gamma \|\Theta\|_2^2$ represents AdamW weight decay regularization ($\gamma = 0.05$).

8 Post-Processing Robustness Defense

To prevent the model from overfitting to pristine uncompressed laboratory datasets, the training pipeline applies on-the-fly forensic augmentations:

- **JPEG Recompression:** Quality factor QF $\in [50, 95]$ to simulate social media compression.
- **Gaussian Blurring:** Kernel $\sigma \in [0.5, 2.0]$ to simulate boundary feathering and anti-forensic smoothing.
- **Additive Sensor Noise:** Gaussian noise $\sigma \in [1.0, 5.0]$ to disrupt naive edge-detection shortcuts.
- **Rescaling:** Random resizing between $0.5\times$ and $1.5\times$ with bilinear interpolation.

9 Benchmark Datasets

The project combines two leading benchmark corpora to ensure balanced multi-class representation:

Dataset	Year	Samples Used	Content Categories	Role in Pipeline
CASIA v1.0/v2.0	2013	≈8,300 Auth / ≈2,800 Spliced	Authentic & Splicing	Provides pristine real samples and splicing manipulations
DEFACTO	2019	8,000 Copy-Move / 8,000 Inpaint	Copy-Move & Inpainting	Provides large-scale copy-move and generative inpainting samples

Table 2: Dataset composition used in Phase 2.

The total corpus comprises **27,061 images** ($8,291$ authentic, $2,770$ splicing, $8,000$ copy-move, and $8,000$ inpainting, capped per class to maintain balance). A held-out test split reserves **2,700 images** for evaluation.

10 Implementation Progress

The core modules of the proposed system are integrated and functional (≈75% completion).

Module / Component	Script (src/)	Implemented Functionality	Status
Feature Extraction	`extractors.py`	2D FFT magnitude, Gaussian HPF kernel, patch slicer	**Integrated**
Tokenization & Fusion	`model.py`	3-stream linear projections & cross-attention fusion layer	**Integrated**
ViT Backbone & Heads	`model.py`	12-layer Transformer, detection & classification heads	**Integrated**
Data Pipeline & Loader	`data_loader.py`	Multi-dataset ingestion and robustness augmentation pipeline	**Integrated**
Training Engine	`train.py`	Masked joint loss loop, AdamW optimizer, checkpointing	**In Progress**
Evaluation & Explainability	`evaluate.py`	Metric reports, confusion matrices, Grad-CAM generator	**Integrated**

Table 3: Current implementation milestone status.

10.1 Functional Verification

Key verified capabilities include:

- End-to-end forward/backward pipeline execution with zero tensor shape or dimensionality mismatch.
- Gated loss training verifying that authentic images do not backpropagate through the classification head.
- Checkpoint management and evaluation harnesses supporting multi-metric confusion matrices.
- Integration of visual explainability using ViT self-attention map extraction and Grad-CAM localization heatmaps.

11 Experimental Results & Evaluation

Preliminary evaluation on the held-out benchmark test set (≈2,700 images) yielded the following metrics:

Binary Forgery Detection

Metric	Score
Accuracy	**88.95%**
Precision	**91.01%**
Recall	**93.29%**
F_1-Score	**92.14%**

Forgery-Type Classification (Macro)

Metric	Score
Accuracy	**67.02%**
Precision	**69.28%**
Recall	**69.73%**
Macro F_1	**69.30%**

Table 4: Preliminary quantitative performance results.

11.1 Error Analysis & Current Limitation

Analysis of the multi-class confusion matrix indicates significant mutual confusion between **Copy-Move** and **Inpainting** categories (305 and 195 misclassifications, respectively). Because inpainting algorithms often blend surrounding textures into the void, the resulting frequency pattern can resemble duplicated intra-image textures. Active optimization is addressing this via hard-negative mining and class-weighted focal loss.

12 Project Evolution & Key Methodological Changes

The methodology underwent significant rigorous corrections during implementation:

1. **Discovery of Mask Leakage (93.74% → 88.95%):** An early experimental setup reported an inflated 93.74% accuracy. Investigation revealed subtle information leakage from ground-truth mask dimensions. Removing mask dependencies corrected the evaluation to a clean, reliable 88.95% accuracy.

2. **Dataset Balancing:** Columbia (363 images) was dropped due to small scale, and IMD2020 was replaced due to format inconsistencies. Merging CASIA and DEFACTO provided a robust 27,061 image corpus.
3. **Class Imbalance Mitigation:** Implemented a `WeightedRandomSampler` to balance gradient updates across under-represented classes (e.g., splicing).
4. **Distributed Multi-GPU Optimization:** Pipeline was optimized for parallel training on $2\times$ NVIDIA T4 GPUs via PyTorch `DataParallel`, achieving stable batch sizes of 64.

13 Design Choices, Trade-Offs & Observations

- **Patch Size (16×16):** While 8×8 patches provide finer boundary resolution, they increase token sequence length to 784, resulting in $16\times$ higher attention computation. 16×16 (196 tokens) achieves the optimal trade-off between localized sensitivity and memory efficiency.
- **Learning Rate (10^{-4}) & Cosine Decay:** Small initial learning rate prevents catastrophic forgetting of pre-trained ViT visual weights, smoothly decaying to zero via Cosine Annealing.
- **Cross-Attention vs. Concatenation:** Cross-attention introduces domain-weighting parameters that allow the model to ignore corrupted RGB streams when images are heavily blurred, falling back onto 2D FFT spectral peaks.

14 Technical Enrichment Activity

To fulfill the individual technical enrichment requirement, all four team members completed specialized professional training:

S.No.	Student Name	Chosen Certification Title	Platform	Hours
1	Dheeraj K B	PyTorch for Deep Learning and Computer Vision	Udemy	14 Hrs
2	Nehan G R M	PyTorch for Deep Learning and Computer Vision	Udemy	14 Hrs
3	Harish Veylan P	PyTorch for Deep Learning and Computer Vision	Udemy	14 Hrs
4	Balaji AR	PyTorch for Deep Learning and Computer Vision	Udemy	14 Hrs

Table 5: Individual Technical Enrichment details.

15 Conclusion & Future Work

This project presents a robust patch-based hierarchical framework for image forgery detection and classification. By tokenizing spatial RGB, 2D FFT spectral magnitude, and high-pass noise residuals as first-class tokens and fusing them via cross-attention, the model overcomes single-domain blindspots. Powered by a Vision Transformer (ViT) backbone, the network leverages global attention to identify both localized splicing cuts and long-range copy-move duplicates.

Immediate future work will focus on:

1. Mitigating Copy-Move and Inpainting confusion using hard-negative mining and class-weighted focal loss.
2. Conducting ablation studies isolating the individual contribution of the frequency and noise streams.
3. Performing rigorous cross-dataset benchmarking against TruFor, MVSS-Net, and ManTra-Net under severe JPEG and blur stress tests.

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